

RGFTSLANet: A Cross-subject Classification Model for decoding EEG-based Motor Imagery tasks

Yun Zhao

School of Smart Health
Chongqing Polytechnic University of
Electronic Technology
Chongqing, China
zhaoyun@cqcet.edu.cn

Bin Jiang

School of Artificial Intelligence
Chongqing University of Technology
Chongqing, China
jb20200132@cqut.edu.cn

Yanying Yan

School of Smart Health
Chongqing Polytechnic University of
Electronic Technology
Chongqing, China
yanyanying@cqcet.edu.cn

Dongyi He

School of Artificial Intelligence
Chongqing University of Technology
Chongqing, China
12223050207@stu.cqut.edu.cn

Xiaoling Zhang

School of Smart Health
Chongqing Polytechnic University of
Electronic Technology
Chongqing, China
zhangxiaoling@cqcet.edu.cn

Shuaidong Zou

School of Smart Health
Chongqing Polytechnic University of
Electronic Technology
Chongqing, China
zoushuaidong@cqcet.edu.cn

Yijie Zhu

School of Smart Health
Chongqing Polytechnic University of
Electronic Technology
Chongqing, China
2369206466@qq.com

Xiyue Hu

College of Computer Science and
Technology
Nanjing University of Aeronautics and
Astronautics
Nanjing, China
1171470872@qq.com

Guanghui Xie

School of Smart Health
Chongqing Polytechnic University of
Electronic Technology
Chongqing, China
xieguanghui@cqcet.edu.cn

Abstract—Motor imagery electroencephalogram (MI-EEG) plays a crucial role in developing brain-computer interfaces (BCIs) that enable natural interaction and autonomous control. Nevertheless, the performance of deep learning models is often hindered by inter-subject variability, which compromises their generalization across different domains. To address this issue, we introduce RGFTSLANet—a novel framework that combines a TSLANet module for signal denoising and temporal pattern extraction, a spatio-temporal convolution component for capturing higher-order features, and a multi-scale Riemannian geometry module to learn subject-invariant distribution characteristics. These complementary features are integrated to improve classification performance. Furthermore, to mitigate inter-subject distribution shifts, RGFTSLANet employs Maximum Mean Discrepancy (MMD) loss for domain adaptation, promoting alignment between source and target feature distributions. Experimental results on the four-class BCI Competition IV 2a dataset demonstrate the model's strong performance, achieving 71.39% accuracy in cross-subject evaluation and 80.71% in single-subject scenarios. Additional ablation and visualization studies confirm RGFTSLANet's capability to reduce cross-subject feature discrepancies and enhance the decoding of MI-EEG signals.

Keywords—motor imagery, brain-computer interfaces; RGFTSLANet, riemannian geometry alignment

I. INTRODUCTION

Brain-computer interface (BCI) technology establishes a direct communication pathway between the human brain and external devices by interpreting electrophysiological brain signals, leading to transformative advances in areas such as medical rehabilitation and intelligent control [1]. Motor imagery electroencephalography (MI-EEG), known for its non-invasive nature and high temporal resolution, is a key signal type in this domain [2]. Decoding methods for MI-EEG

are generally divided into traditional techniques and deep learning approaches. Traditional methods depend heavily on manual feature engineering and shallow machine learning models, which often fail to model complex nonlinear patterns in EEG data [3]. In contrast, deep learning—especially convolutional neural networks (CNNs)—has gained broad adoption in MI-EEG decoding due to its powerful nonlinear modeling capacity. CNNs can automatically learn spatiotemporal or latent features from EEG signals, substantially improving decoding accuracy in certain scenarios [4].

However, most current studies focus on subject-dependent models, leaving a critical gap in subject-independent research, which is essential for enhancing model generalizability and enabling real-world BCI deployment. Moreover, variability in recording devices and environmental noise further complicates cross-subject EEG analysis. Existing models often fall short in generalization because they do not adequately capture intra- and inter-subject feature relationships [5].

To bridge this gap, transfer learning strategies have been introduced to improve cross-subject MI-EEG decoding. For example, Zhang et al. developed an adaptive transfer learning approach that fine-tunes pre-trained deep CNN models for new subjects, thereby boosting cross-subject performance [6]. Similarly, semi-supervised multi-source transfer learning models have shown promise by learning informative and domain-invariant representations [7]. Despite these advances, current algorithms still underperform in fully capturing domain-invariant features and often overlook the noise sensitivity of cross-subject EEG, which constrains decoding accuracy.

In response, this paper proposes RGFTSLANet (Riemannian MMD-enhanced EEG TSLANet), a novel neural network designed to address these challenges. The model first employs a Time Series Lightweight Adaptive Network (TSLANet) to denoise EEG signals and capture complex temporal dependencies, enabling preliminary signal decoding. Next, a dedicated spatio-temporal convolution module extracts higher-order spatio-temporal features. Additionally, a multi-scale Riemannian geometry feature module is incorporated to learn domain-invariant distribution characteristics across subjects. These features are fused with those from the spatio-temporal module to enhance classification performance. To further mitigate inter-subject distribution shifts, a domain adaptation mechanism based on Maximum Mean Discrepancy (MMD) loss is integrated during training, aligning feature distributions between source and target domains. This design reduces the impact of individual variability and enables effective cross-subject MI-EEG decoding.

II. METHODS

A. RGFTSLANet model

A novel deep learning network, RGFTSLANet, was designed as a transfer learning framework that facilitates feature transfer from the source domain to the target domain through a tailored model architecture. As illustrated in Figure 1, RGFTSLANet comprises three core components: (1) the TSLANet module, which learns temporal patterns from EEG signals; (2) a spatio-temporal convolution module that extracts time-frequency features; and (3) a multi-scale Riemannian convolution module dedicated to capturing cross-subject Riemannian geometric characteristics.

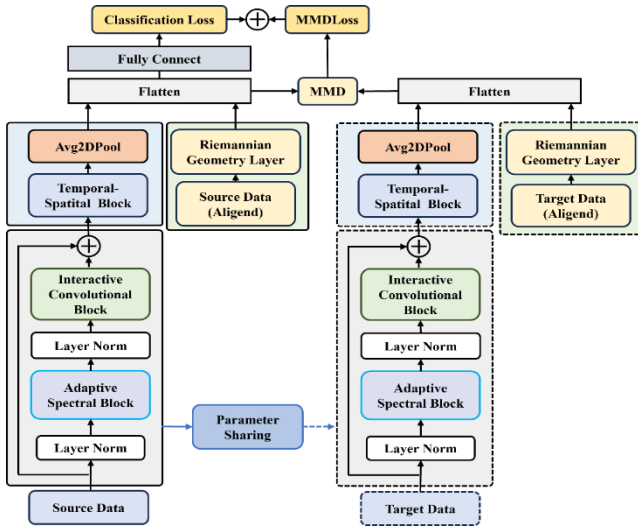


Figure 1. The architecture of RGFTSLANet

The model accepts two types of input: preprocessed raw EEG data and Riemannian geometric features derived from both source and target domains. It processes these inputs through dual branches. The left branch takes raw EEG signals and employs TSLANet to learn temporal representations, which are further refined via temporal and depth-wise convolutions within the spatio-temporal module. The right branch processes Riemannian features to learn domain-invariant distribution representations across subjects. The outputs from both branches are then fused and passed through a fully connected layer with a SoftMax classifier to generate probability estimates for MI-EEG categories.

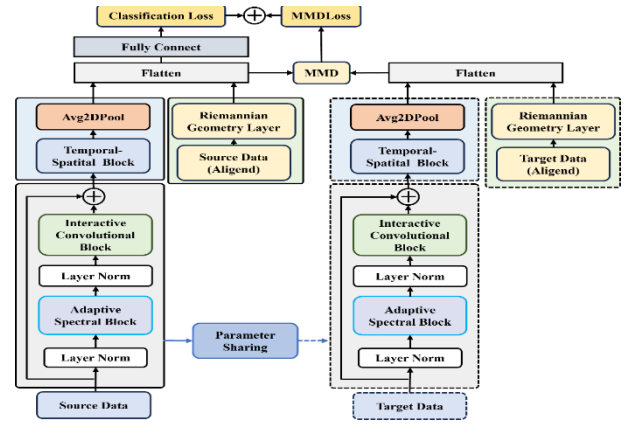


Figure 2. The training process of RGFTSLANet

A key aspect of the model is its incorporation of Maximum Mean Discrepancy loss (MMDLoss) for domain alignment, as depicted in Figure 2. Domain adaptation in this context aims to transfer knowledge from the source domain to related target domains by identifying a transformation that minimizes distribution discrepancies. MMD operates by embedding both source and target domain data into a reproducing kernel Hilbert space (RKHS) and computing the distance between their mean embeddings. This distance quantifies inter-domain divergence and is incorporated as a regularization term in the overall loss function. By aligning global feature distributions through shared network parameters, the model effectively learns cross-domain invariant features.

B. Experimental Setup

This study evaluates the classification performance of the proposed model for both within-subject and cross-subject motor imagery EEG (MI-EEG) recognition using a publicly available dataset (see Table 1). We compare our approach against several state-of-the-art methods and conduct ablation studies to demonstrate its contribution.

Table 1 The summary of the BCI Competition IV dataset 2a

Dataset	Subjects	Channels	Classes	Trials	Samples	Sampling Rate
BCICIV-2a [31]	9	22	4	288*9	1000	250

The BCI Competition IV dataset 2a [8], a widely adopted benchmark provided by Graz University of Technology was selected for MI-EEG classification. This dataset contains EEG recordings from nine subjects performing four motor imagery tasks: left hand, right hand, both feet, and tongue. Signals were captured using 22 Ag/AgCl electrodes sampled at 250 Hz. Each subject participated in two sessions conducted on different days, with each session comprising 288 trials (72 trials per class).

As previously mentioned, covariance matrices are computed separately from the training and validation sets. Riemannian alignment is then applied to these matrices to reduce inter-subject variability and improve model generalization. Consequently, input to the model in cross-subject experiments includes both raw EEG data and the aligned Riemannian features. For cross-subject evaluation, we adopt the leave-one-subject-out (LOSO) strategy: one subject's data serves as the test set, while the remaining subjects' data are used for training.

To assess model performance, we employ two common evaluation metrics in MI-EEG classification: accuracy (ACC) and kappa coefficient (KAPPA). Accuracy measures the proportion of correctly classified samples, while kappa quantifies classification agreement beyond chance.

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

where, TP , TN , FP , and FN represent the true positives, true negatives, false positives, and false negatives, respectively.

$$\text{KAPPA} = \frac{P_0 - P_e}{1 - P_e} \quad (2)$$

The proposed model is implemented in Python 3.8 using PyTorch and trained on a GeForce RTX 3080Ti GPU. We use a batch size of 64 and train for 500 epochs, optimized by Adam with a learning rate of 0.0001 and weight decay of 0.075. The TSLANet module contains three layers, and L2 regularization is applied to prevent overfitting. Throughout training, the model weights corresponding to the lowest validation loss are saved and subsequently used for testing.

Table 2 Single-subject classification results

Methods	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg. (kappa)
EEGNet [9]	84.34	54.06	87.54	63.59	67.39	54.88	88.80	76.75	74.24	72.40 (0.63)
EEG-TCNet [10]	85.77	65.02	94.51	64.91	75.00	61.40	87.36	83.76	78.03	77.35 (0.71)
EEG-ITNet [11]	84.38	62.85	89.93	69.10	74.31	57.64	88.54	83.68	80.21	76.74 (0.71)
LMDANet [12]	86.50	67.40	91.70	77.40	65.60	61.10	91.30	83.30	85.40	78.80 (0.71)
RGFTSLANet (ours)	88.31	59.88	95.56	80.20	71.59	62.31	96.33	85.38	87.40	80.78 (0.76)

B. Cross-subject classification

Table 3 further summarizes the cross-subject classification results, where RGFTSLANet attained an average accuracy of 71.39% using fixed parameters and hyperparameters across all test subjects. This represents a notable improvement of 21.4% over the classical EEGNet (50.60%). The results reveal clear limitations in the cross-subject generalization capability of conventional deep learning models. Although these methods exhibit strong representational power, their restricted receptive fields and

inability to explicitly model inter-subject distribution shifts lead to suboptimal decoding performance, even in subject-dependent scenarios. More critically, they do not account for domain-invariant feature learning. While performing well in the source domain, such models fail to align feature distributions across different subjects, resulting in degraded cross-subject classification. These findings further validate the effectiveness of the proposed model in aligning feature distributions between source and target domains, enhancing adaptive learning, and improving cross-subject MI-EEG decoding performance.

Table 3 Cross-subject classification results

Methods	S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg. (kappa)
EEGNet [9]	65.53	54.06	73.42	53.10	61.04	49.69	66.87	63.12	74.24	50.60 (0.27)
ShallowConvNet [3]	70.51	46.25	71.22	50.65	62.44	49.07	64.72	67.30	62.53	60.45 (0.35)
EEG-TCNet [10]	67.40	62.94	79.90	49.44	60.00	51.60	71.52	71.11	69.12	62.35 (0.45)
EEG-ITNet [11]	68.13	56.30	75.87	55.21	65.40	56.89	74.30	67.66	67.00	63.65 (0.46)
Transformer-Based [13]	61.61	59.12	64.35	62.36	67.35	63.02	65.62	63.67	64.97	63.56 (0.46)
RGFTSLANet (ours)	76.01	49.79	84.58	64.93	67.78	61.18	80.00	80.73	77.54	71.39 (0.59)

C. Ablation Study

To evaluate the contribution of the two cross-domain learning components—the multi-scale Riemannian feature module and the MMD loss—to cross-subject performance, we conducted a series of ablation experiments across two datasets.

III. RESULTS AND DISCUSSION

A. Single-subject classification

As presented in Table 2, this study compares the classification accuracy and Kappa values of the proposed model against several advanced deep learning methods on the BCICIV-2a dataset under a subject-specific training scheme. By assessing the feature extraction capabilities of these models in within-subject experiments—where the MMDLoss and Riemannian feature fusion components were removed from our framework—the base network still exhibited consistent advantages. Specifically, the proposed model outperformed classical approaches such as EEGNet and EEG-ITNet by 8.38% and 4.04%, respectively, highlighting the limitations of these conventional methods in multi-class MI-EEG tasks. Moreover, our model surpassed temporal-based architectures, exceeding EEG-TCNet by 3.43%. It also achieved competitive performance against recent state-of-the-art models, including Transformer-based frameworks and LAMNet—a method specifically tailored for EEG processing—reaching 80.78% accuracy in subject-specific evaluation, which is approximately 1.5% higher than other leading deep learning models.

The results are summarized in Table 4. Four model configurations were compared: the full model, the model with both components removed, the model without the Riemannian module, and the model without MMD loss.

Through systematic ablation and comparative analysis, we identified a synergistic effect between the Riemannian feature module and the MMD-based domain adaptation mechanism in enhancing motor imagery EEG classification. The performance gains on the BCICIV-2a dataset can be attributed to three key innovations. First, the multi-scale Riemannian module improves cross-subject alignment by learning geometric structure from covariance matrices, contributing a

0.16% accuracy gain when used independently. Second, the MMD loss reduces inter-subject domain shift by minimizing distribution discrepancy between source and target domains, yielding a 1.42% improvement when applied alone. Most notably, the combined use of both components produces a compound effect, leading to a 4.4% performance increase on the BCICIV-2a dataset, which confirms their complementary roles in cross-subject generalization.

Table 4 Ablation study results

Methods		S01	S02	S03	S04	S05	S06	S07	S08	S09	Avg. (kappa)
Rieman	MMDLoss										
×	×	74.28	51.23	81.34	58.30	56.32	55.42	76.56	77.41	73.09	66.99 (0.55)
×	√	75.22	49.16	82.41	61.02	66.26	59.56	78.24	80.74	73.04	69.40 (0.58)
√	×	73.68	49.01	80.91	61.63	57.33	54.26	75.56	80.42	71.52	67.15 (0.55)
√	√	76.01	49.79	84.58	64.93	67.78	61.18	80.00	80.73	77.54	71.39 (0.59)

Deeper analysis indicates that the TSLanet module improves noise robustness and feature representation by capturing both long- and short-term temporal dependencies in EEG signals, combined with spatio-temporal convolutional networks that model higher-order spatio-temporal features. By integrating this hierarchical feature extraction architecture with multi-scale Riemannian geometric features, the model effectively captures both local dynamic properties and global statistical characteristics of EEG data. In terms of cross-subject generalization, the end-to-end domain adaptation mechanism based on MMDLoss not only mitigates inter-subject distribution discrepancies but also promotes the learning of more discriminative domain-invariant features. Experimental results confirm that the core strengths of RGFTSLanet originate from: the stable geometric representations enabled by multi-scale Riemannian features, the discriminative spatio-temporal patterns extracted by TSLanet and spatio-temporal convolutions, and the domain adaptation strategy driven by MMDLoss that optimizes cross-subject distribution alignment. Ultimately, the model achieves state-of-the-art classification performance on two public datasets, offering a robust and generalizable solution for practical motor imagery brain-computer interface applications.

ACKNOWLEDGMENT

This research was funded by Natural Science Foundation of Chongqing (CSTB2025NSCQ-GPX0794), and the Scientific and Technological Research Program of the Chongqing Education Commission (KJZD-K202303103).

REFERENCES

- [1] D. J. McFarland, J. R. Wolpaw, Brain-computer interface use is a skill that user and system acquire together, *PLoS biology* 16 (2018) e2006719.
- [2] K. P. Thomas, C. Guan, C. T. Lau, A. P. Vinod, K. K. Ang, A new discriminative common spatial pattern method for motor imagery brain-computer interfaces, *IEEE transactions on biomedical engineering* 56 (2009) 2730–2733.

- [3] R. Tibor Schirrmester, L. Gemein, K. Eggenberger, F. Hutter, T. Ball, Deep learning with convolutional neural networks for decoding and visualization of eeg pathology, *arXiv e-prints* (2017) arXiv-1708.
- [4] A. Craik, Y. He, J. L. Contreras-Vidal, Deep learning for electroencephalogram (eeg) classification tasks: a review, *Journal of neural engineering* 16 (2019) 031001.
- [5] Y. Chen, R. Yang, M. Huang, Z. Wang, X. Liu, Single-source to single-target cross-subject motor imagery classification based on multisub-domain adaptation network, *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 30 (2022) 1992–2002.
- [6] K. Zhang, N. Robinson, S.-W. Lee, C. Guan, Adaptive transfer learning for eeg motor imagery classification with deep convolutional neural network, *Neural Networks* 136 (2021) 1–10.
- [7] F. Zhang, H. Wu, Y. Guo, Semi-supervised multi-source transfer learning for cross-subject eeg motor imagery classification, *Medical & Biological Engineering & Computing* 62 (2024) 1655–1672.
- [8] C. Brunner, R. Leeb, G. Müller-Putz, A. Schlögl, G. Pfurtscheller, Bci competition 2008–graz data set a, *Institute for knowledge discovery (laboratory of brain-computer interfaces), Graz University of Technology* 16 (2008) 34.
- [9] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, B. J. Lance, Eegnet: a compact convolutional neural network for eeg-based brain-computer interfaces, *Journal of neural engineering* 15 (2018) 056013.
- [10] T. M. Ingolfsson, M. Hersche, X. Wang, N. Kobayashi, L. Cavigelli, L. Benini, Eeg-tenet: An accurate temporal convolutional network for embedded motor-imagery brain-machine interfaces, in: *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC), IEEE, 2020*, pp. 2958–2965.
- [11] A. Salami, J. Andreu-Perez, H. Gillmeister, Eeg-itnet: An explainable inception temporal convolutional network for motor imagery classification, *IEEE Access* 10 (2022) 36672–36685.
- [12] Z. Miao, M. Zhao, X. Zhang, D. Ming, Lmda-net: A lightweight multi-dimensional attention network for general eeg-based brain-computer interfaces and interpretability, *NeuroImage* 276 (2023) 120209.
- [13] X. Tan, D. Wang, J. Chen, M. Xu, Transformer-based network with optimization for cross-subject motor imagery identification, *Bioengineering* 10 (2023) 609.